Activity 16

Writing a C program based upon user choice. It involves construction of functions and using them in choice cases of int main.

1. Check if all digits are in increasing order.
2. Check if all digits are in decreasing order.
3. Check the largest and smallest digit of that number.
4. Check the second smallest and second largest number.

Program

```c
#include <stdio.h>

void increasingorder(){
	int num, prev, digit;
	printf("Enter a positive integer: ");
	scanf("%d", &num);
	prev = num % 10;
	num /= 10;
	int flag = 1;
	while(num > 0){
		digit = num % 10;
		if(digit >= prev){
			flag = 0;
			break;
		}
		prev = digit;
		num /= 10;
	}
	if(flag) printf("digits are in increasing order.\n");
	else printf("digits are not in increasing order.\n");
}

void decreasingorder(){
	int num, prev, digit;
	printf("Enter a positive integer: ");
	scanf("%d", &num);
	prev = num % 10;
	num /= 10;
	int flag = 1;
	while(num > 0){
		digit = num % 10;
		if(digit <= prev){
			flag = 0;
			break;
		}
		prev = digit;
		num /= 10;
	}
	if(flag) printf("digits are in decreasing order.\n");
	else printf("digits are not in decreasing order.\n");
}

void largestsmallest(){
	int num, digit;
	printf("enter a positive integer: ");
	scanf("%d", &num);
	int largest = 0, smallest = 9;
	while(num > 0){
		digit = num % 10;
		if(digit > largest) largest = digit;
```

```c
                digit = num % 10;
                if(digit > largest) largest = digit;
                if(digit < smallest) smallest = digit;
                num /= 10;
        }
        printf("Largest digit: %d, smallest digit: %d\n", largest, smallest);
}

void secondlargestsmallest(){
        int num, digit;
        printf("Enter a positive integer: ");
        scanf("%d", &num);
        int freq[10] = {0};
        while(num > 0){
                digit = num % 10;
                freq[digit]++;
                num /= 10;
        }
        int largest=-1, secondlargest=-1, smallest=10, secondsmallest=10;
        for(int i=9; i>=0; i--){
                if(freq[i] > 0){
                        if(largest == -1) largest = i;
                        else if(secondlargest == -1) { secondlargest = i; break; }
                }
        }
        for(int i=0; i <= 9; i++){
                if(freq[i] > 0){
                        if(smallest == 10) smallest = i;
                        else if(secondsmallest == 10) { secondsmallest = i; break;}
                }
        }
        if(secondlargest == -1 || secondlargest == 9)
                printf("Secondlargest or smallest digit not found.\n");
        else
                printf("Secondlargest = %d,  secondsmallest = %d", secondlargest, secondsmallest);
}

int main(){
        int choice;
        do {
                printf("\n ----Menu---- \n");
                printf("1. Check if digits are in increasing order.\n");
                printf("2. Check if digits are in decreasing order.\n");
                printf("3. find largest and smallest digit.\n");
                printf("4. find second largest and second smallest digit.\n");
                printf("5. Exit.\n");
                printf("Enter your choice :");
                scanf("%d", &choice);

                switch(choice){
```

```c
	while(num > 0){
		digit = num % 10;
		freq[digit]++;
		num /= 10;
	}
	int largest=-1, secondlargest=-1, smallest=10, secondsmallest=10;
	for(int i=9; i>=0; i--){
		if(freq[i] > 0){
			if(largest == -1) largest = i;
			else if(secondlargest == -1) { secondlargest = i; break; }
		}
	}
	for(int i=0; i <= 9; i++){
		if(freq[i] > 0){
			if(smallest == 10) smallest = i;
			else if(secondsmallest == 10) { secondsmallest = i; break;}
		}
	}
	if(secondlargest == -1 || secondlargest == 9)
		printf("Secondlargest or smallest digit not found.\n");
	else
		printf("Secondlargest = %d,  secondsmallest = %d", secondlargest, secondsmallest);
}

int main(){
	int choice;
	do {
		printf("\n ---Menu--- \n");
		printf("1. Check if digits are in increasing order.\n");
		printf("2. Check if digits are in decreasing order.\n");
		printf("3. find largest and smallest digit.\n");
		printf("4. find second largest and second smallest digit.\n");
		printf("5. Exit.\n");
		printf("Enter your choice :");
		scanf("%d", &choice);

		switch(choice){
			case 1: increasingorder(); break;
			case 2: decreasingorder(); break;
			case 3: largestsmallest(); break;
			case 4: secondlargestsmallest();break;
			case 5: printf("Exitingprogram.\n"); break;
			default: printf("Invalid choice. Try again.\n");
		}
	}while(choice != 5);
	return 0;
}
```

Algorithm

1.START

2.void func1 for checking if digits are in increasing order or not.

3.void func2 for checking if digits Are in decreasing order or not.

4.void func3 for checking the largest and smallest digit of a number.

5.void func4 for checking the second largest and second smallest digit.

6.using switch builtin to demonstrating the choice to user and using case with funcs just made and printing result.

7.STOP

Test case table for num 123456

Input choice	Expected output	Actual output	Result
1.	yes	yes	Pass
2.	no	no	Pass
3.	1, 6	1,6	Pass
4.	2, 5	2, 5	Pass